For my software design and engineering artifact, I chose my Animal Shelter CRUD project form CS 340. I originally created this project during the course using Python and MongoDB. The program uses an AnimalShelter class to perform the four main CRUD operations: create, read, update, and delete. I chose this artifact because it demonstrates my experience working with Python, databases, and software development.

For my enhancement, I focused on making the original program safer and easier to maintain. I added input validation to each CRUD method so the program can better handle incorrect information. I also prevented empty queries from being used for update and delete operations. This will help avoid accidentally changing or deleting multiple records. I improved the error handling by using PyMongoError instead of a general Exception and added docstrings to make the code easier to understand.

These changes helped me make progress toward the course outcomes I planned to demonstrate. The enhancement shows my ability to improve an existing program and make better design based on weaknesses in the original code. It also improves the reliability and security of the program.

One challenge I faced was deciding how much to change without completely replacing the original project. I wanted the enhanced version to still be recognizable while showing the improvements. After making these changes, I tested the new validation and confirmed that the program correctly handled invalid create, read, update, and delete requests.